Math 330

Matthew Murillo

Fall Semester 2022

This document covers lecture, textbook, and video notes for Yale's Math 330 course taught by Sekhar Tatikonda. These are not the official class notes and therefore have not been reviewed by the professor. There could be information missing or incorrect so please be sure to validate with other sources when using.

Contents

I Lecture Notes **2**

1 Introduction to the course **3**
1.1 Measure Theoretic Definitions . . . 3
1.2 Problem with the Reimann integral . . . 3
1.3 Defining a measure . . . 5

2 Rings, Algebras, and Sigma Algebras **7**
2.1 Defining measures . . . 7
2.2 Families of sets . . . 7
2.3 Generating sets . . . 8

3 Sigma Algebras and Outer Measures **10**
3.1 Motivation . . . 10
3.2 Generating larger families of sets . . . 10

4 Carath/eodory's Extension Theorem **13**
4.1 Construction of an outer-measure . . . 13
4.2 Carath/eodory's Extension Theorem . . . 14
4.3 Existence and Uniqueness . . . 16

Part I
Lecture Notes

1 Introduction to the course

This a measure theoretic course of probability. This means that we will go over proofs for classic probability notions from a measure theoretic perspective. Such ideas include the Law of Large Numbers, the Central Limit Theorem, conditions, etc.

We should have a general idea as to what these ideas imply, but this course will go into greater detail on their foundations. The class covers two textbooks which we will also cover in these notes: Pollard and Durrett.

This course assumes knowledge on probability, linear algebra, multivariable calculus, and real analysis.

1.1 Measure Theoretic Definitions

Definition 1. Measure Space

A measure space is defined by the following triple $(\Omega, \mathcal{F}, \mu)$ where Ω is the outcome space, $\mathcal{F}$ is a σ-filed, and μ is a measure. The measure space is the the central mathematical object of probability theory. In the first section of the course, we will spend time developing a measure theoretic notion of σ-field and measure more generally. We will then apply these ideas to probability.

Example. Ω Sample/outcome space

Definition 2. Ω is the set from which our family of sets will be constructed. In probability theory, you can think of Ω as Nature which contains the set of all possible events.

Example. $(0, 1)$, $\mathbb{R}^n$, $\{0, 1, 2, 3, 4, ..., n\}$, etc

Definition 3. $\mathcal{F}$ σ-field, (σ-algebra)

A σ-field of measurable sets, measurable meaning we can define a measure on the set, is a set of sets, or a family of sets such that $\mathcal{F} \subseteq 2^{\Omega}$. $\mathcal{F}$ has more defining properties which we will cover in greater detail later. In probability, think of the sets in our σ-field $\mathcal{F}$ as the events which we observe.

We also call $(\Omega, \mathcal{F})$ a measurable space. Given Ω and $\mathcal{F}$ we can now define a measure μ

Definition 4. $\mu : \mathcal{F} \to \mathbb{R}^+$ is a measure

We call μ probability measure if $\mu(\Omega) = 1$. Essentially, μ is a function that takes elements from our σ-algebra as inputs and maps them to the (typically) extended real line, $\mathbb{R}^+$.

In this lecture we will cover where we might run into problems when defining attempting to define valid measures. Let's first take a look at something that should be familiar to us – the Reimann integral.

1.2 Problem with the Reimann integral

Recall the Reimann integral from calculus. We define the Reimann integral as

$$\int_a^b f(x)\, dx$$

Where $\Pi = \{\pi_i\}$ is the set of disjoint intervals that partition our interval $[a, b]$. Furthermore, for each partition, we can define π_i, we define a_i to be the infimum of $f(x)$ evaluated within that interval and b_i as the supremum. More formally stated, we have that

$$a_i = \inf_{x \in \pi_i} f(x)$$
$$b_i = \sup_{x \in \pi_i} f(x)$$

Moreover, we define the lower bound (LB) and upper bound (UB) given our set of intervals as

$$LB(\Pi) \equiv \sum_i a_i \cdot |\pi_i|$$

$$UB\left(\Pi\right) \equiv \sum_i b_i \cdot \left|\pi_i\right|$$

and we set that f is a Reimann integral when $\sup_\Pi LB\left(\Pi\right) = \inf_\Pi UB\left(\Pi\right)$. The following two figures depicts the graphs for the LB vs the UB. The idea behind the Reimann integral is that as the number of partitions increases, the size of our LB increases and the size of our UB decreases until the two are equal.

Figure 1.1: Lower Bound

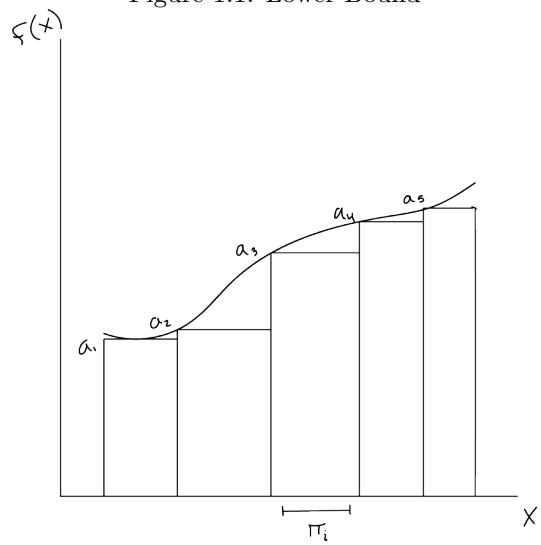
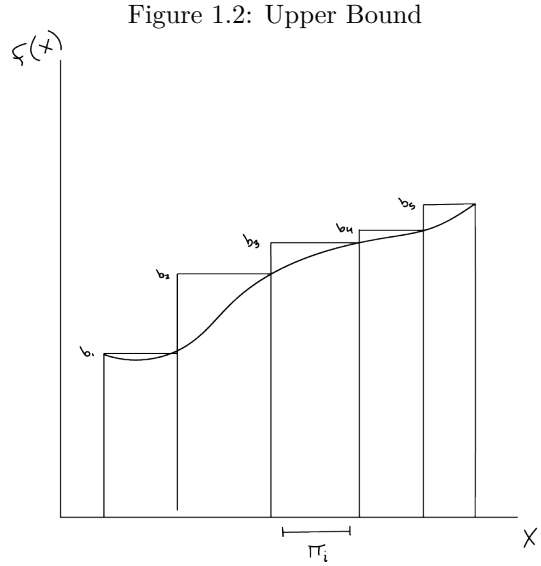


Figure 1.2: Upper Bound



Theorem 1. *Riemann Integrability.*
f is Riemann integrable if

1. *f is bounded*
2. *Set of discontinuities has Lebesgue measure equal to 0, i.e., there are a countable number of them.*

There are several examples where the Riemann integral breaks. A class example is the Dirichlet function.

$$f : [0,1] \to \{0,1\} : \begin{cases} 0 & x \in \mathbb{Q}^c \\ 1 & x \in \mathbb{Q} \end{cases}$$

In this example, f is discontinuous everywhere. For any partition you take, the $LB = 0$ and the $UP = 1 \therefore$ they will never equal.

The two main problems with the Riemann integral can be summarized as follows:

1. Holds for a limited class of functions
2. Basic limit operations don't hold

Example. Let's order the rational $a_1, a_2, ...$ Now let's define the function

$$f_n(x) = \begin{cases} 1 & x \in \{a_1, \ldots, a_n\} \\ 0 & \text{else} \end{cases}$$

In this example, $f_n(x) \to f(x)$ point-wise where $f(x)$ is the Dirichlet function. But,

$$\lim_n \int f_n(x)\, dx \not\to_? \int f(x)\, dx$$

does not exist.

1.3 Defining a measure

Formally, we defined a measure as a function from a family of sets to the real line (or extended real line). Let's consider the following example

$$\mu : 2^\Omega \to \mathbb{R}^+$$

where μ is a measure from the power set to the extended reals. For our measure to be well-defined, we want that at least the following three properties hold for $A, B \subset \Omega$.

1. $\mu(\emptyset) = 0$. The measure of the empty set is equal to 0
2. If $A \cap B = \emptyset \Rightarrow \mu(A \cup B) = \mu(A) + \mu(B)$. If A and B are disjoint, then the measure of their union is equal to the sum of the measures of each.
3. If $A \subseteq B \Rightarrow \mu(A) \leq \mu(B)$. If A is contained in B then the measure of A is less than or equal to the measure of B. This is called monotonicity.

However, these definitions are not sufficient to construct a well-defined measure. Let's consider the following counterexample.

Example. Let $p \neq q \in \Omega$ and $A \subseteq 2^\Omega$. Now, let's define our "measure"

$$\mu(A) = \begin{cases} 1 & \text{if } p, q \in A \\ 0 & \text{else} \end{cases}$$

Then, we see that $\mu(\{p\}) + \mu(\{q\}) = 0 + 0 \neq 1 = \mu(\{p, q\})$.

This result implies that μ cannot be any arbitrary function from the power-set.

Let's take an even closer look using a more common measure, the Lebesgue measure. We define the Lebesgue measure as

$$\mu_L([a, b]) = b - a$$

where $[a, b] \subseteq \mathbb{R}$. Let's consider a simple example where our loose definition of a measure fails using a nice set and a common measure

Example. Let $\Omega = [0, 1]$, $A_i = \{i\}$ for $i \in [0, 1]$. Clearly we have that $\mu_L(A_i) = 0$ and $\cup A_i = [0, 1]$. However, we see that property (2) of our definition fails

$$1 = \mu_L([0, 1]) = \mu_L\left(\cup_{i \in [0,1]} A_i\right) \neq \sum_{i \in I} \mu_L(A_i) = 0$$

The problem here being that $i \in I$ is uncountable.

To summarize, not all sets are measurable, and even if they are measurable we cannot deal with uncountable additivity. Therefore, throughout this course we will be dealing with countable additivity of measurable sets.

2 Rings, Algebras, and Sigma Algebras

In this lecture, we introduce the sets for which can be constructed and/or for where they are well defined. We formally define what it means to be a measure and study properties for the families of sets which we will be using throughout the course.

2.1 Defining measures

In the previous lecture, we gave a very loose definition for measures, and we saw examples where this definition failed. Now, we will formally define what it means to be a measure.

Definition 5. A set function $\mu : \mathcal{A} \to [0, \infty]$ is a countably (also implying finitely) additive measure with respect to $\mathcal{A}$ if

1. $\emptyset \in \mathcal{A}$ then $\mu(\emptyset) = 0$
2. If $A_1, \ldots, A_n \in \mathcal{A}$ and $\cup_{i=1}^n A_i \in \mathcal{A}$ are pairwise disjoint, then $\mu(\cup_{i=1}^n A_i) = \sum_{i=1}^n \mu(A_i)$. Finite additivity.
3. If $\{A_i\}_{i \in I} \subseteq \mathcal{A}$ are disjoint and I is a countable index set and $\cup_{i \in I} A_i \in \mathcal{A}$, then $\mu(\cup_{i \in I} A_i) = \sum_{i \in I} \mu(A_i)$.

You might be wondering about uncountably additivity and monotonicity. Recall from our final example in the previous lecture with the Lebesgue measure – our definition fails for when I is uncountable. And as for monotonicity, it is implied by finite additivity.

Proposition 1. *Finitely additive measures are monotone.*

Proof. If $A, B, B \backslash A \in \mathcal{A}$ and $A \subseteq B$ then $\mu(A) \leq \mu(B)$. We can say $\mu(B) = \mu(A \cup B \backslash A) = \mu(A) + (B \backslash A) \geq \mu(A) \therefore \mu(B) \geq \mu(A)$ □

Corollary 1. *Countably additive measures are also monotone*

Using similar techniques, we can also show that finitely additive measure are also finitely sub-additive

Proposition 2. *Finitely additive measures are finitely sub-additive.*

Proof. WTS $\mu(A \cup B) \leq \mu(A) + (B)$ provided that $A, B, A \cup B, B \backslash A \in \mathcal{A}$. We can say $\mu(A \cup B) = \mu(A \cup B \backslash A) = \mu(A) + \mu(B \backslash A) \leq \mu(A) + \mu(B) \therefore \mu(A \cup B) \leq \mu(A) + \mu(B)$. □

A special type of measure which we will see throughout the course is a probability measure.

Definition 6. A measure is a probability measure if $\Omega \in \mathcal{A}$ and $\mu(\Omega) = 1$.

To understand when such measures exist, we need to examine the families of sets measures are built on.

2.2 Families of sets

All the definitions of measures that we've given require certain closure properties of the set $\mathcal{A}$. In this section, we will give more structure to $\mathcal{A}$, and our end goal will be to build a measure by taking advantage of the composition rules of $\mathcal{A}$.

Definition 7. $\mathcal{A} \subseteq 2^\Omega$ is a ring if

1. $\emptyset \in \mathcal{A}$
2. $A, B \in \mathcal{A} \Rightarrow A \cup B \in \mathcal{A}$
3. $A, B \in \mathcal{A} \Rightarrow A \backslash B \in \mathcal{A}$

Notice that $A \cap B = A \backslash (A \backslash B) \in \mathcal{A}$. Therefore, by induction we can say that rings are closed under finite $\cup, \cap$, and relative complements.

Definition 8. $\mathcal{A}$ is an algebra/field if $\mathcal{A}$ is a ring such that $\Omega \in \mathcal{A}$.

Since $\mathcal{A}$ is a ring, we know it must also be closed under finite set operations

Definition 9. $\mathcal{A}$ is a σ-ring if $\mathcal{A}$ is a ring that is closed under countable unions.

Since $\mathcal{A}$ is a ring and we know rings are closed under relative complements, then we also know that σ-rings must also be closed under countable intersections. This is because $\cap_{n=1}^{\infty} A_n = A_1 \setminus (\cup_{n=1}^{\infty} A_1 \setminus A_n)$

Definition 10. $\mathcal{A}$ is a σ-algebra/σ-field if $\mathcal{A}$ is an algebra closed under countable unions

Definition 11. $\mathcal{A}$ is a semi-ring if

1. $\emptyset \in \mathcal{A}$
2. $A, B \in \mathcal{A} \Rightarrow A \cap B \in \mathcal{A}$
3. $A \setminus B = \cup_{1 \leq j \leq n} C_j$, where $C_j \in \mathcal{A}$ are pairwise disjoint

2.3 Generating sets

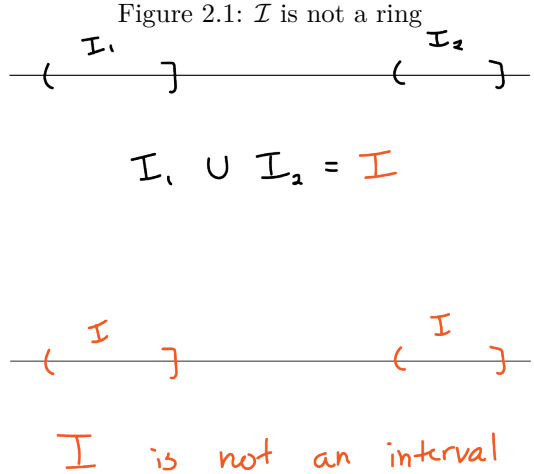
Now, let's discuss how we can generate families of sets from smaller families of sets. We'll first consider the set

$$\mathcal{I} = \{(a, b]\}$$

defined to be the set of all intervals in $\mathbb{R}$.

The first thing to notice is that $\mathcal{I}$ is not a ring. Consider the union of two disjoint $I_1, I_2 \in \mathcal{I}$. Their union is not an interval, rather it is a finite union of disjoint elements from $\mathcal{I}$.

Figure 2.1: $\mathcal{I}$ is not a ring



However, we do know that $\mathcal{I}$ is a semi-ring. In this example above, we can say that $\mathcal{I}$ is a semi-ring because it satisfies our definition of "relative complements".

Notice that we can easily define our Lebesgue "measure" on the semi-ring $\mathcal{I}$. We can just say that $\mu_{L_0}((a, b]) = b - a$. Our goal is to construct a ring from the semi-ring, extending our notion of the Lebesgue measure such that $\mu_{L_1}|_{\mathcal{I}} = \mu_{L_0}$.

Theorem 2. *For any semi-ring, $\mathcal{A}$, let $\mathcal{B}$ be the set of all finite disjoint unions of sets in $\mathcal{A}$. Then $\mathcal{B}$ is a ring.*

Proof. We know that $\mathcal{B}$ is a ring because it satisfies the definition

1. $\emptyset \in \mathcal{B}$
2. $A = \cup_{i=1}^{n} A_i$ disjoint, $B = \cup_{j=1}^{n} B_j$ disjoint where $A_i, B_i \in \mathcal{A}$, then by definition $A, B \in \mathcal{B}$. However, we can also say that $A \cap B = \cup_{i.j} A_i \cap B_j$ which is the union of finite, disjoint intervals so $A \cap B \in \mathcal{B}$.
3. $B \backslash A \in \mathcal{B}$. We can say that $B \backslash A = \cup_{i,j} B_j \backslash A_i$ where $B_j \backslash A_i$ is disjoint so we're done.
4. $A \cup B = A \cup (B \backslash A)$ is disjoint where $A \in \mathcal{B}, B \backslash A \in \mathcal{B}$ so their union is also in $\mathcal{B}$

□

Let's now define the set $\mathcal{J}$ to be the ring generated by the finite disjoint union of elements in $\mathcal{I}$. Now, let's extend μ_{L_0} on our semi-ring $\mathcal{I}$ to μ_{L_1} defined on our ring $\mathcal{J}$. If $J \in \mathcal{J}$ then we know that $J = \cup_{i=1}^{n} I_i$ where $I_i \in \mathcal{I}$ are disjoint. So let's define μ_{L_1} as

$$\mu_{L_1}(J) \triangleq \sum_{i=1}^{n} \mu_{L_0}(I_i)$$

Theorem 3. *μ_{L_1} on $\mathcal{J}$ is a finitely additive measure.*

Proof. We need to check that μ_{L_1}satisfies the conditions for what it means to be a finitely additive measure on a set.

1. $\mu_{L_1}(\emptyset) = 0$
2. $J_1, J_2 \in \mathcal{J}$ disjoint, then we have $J_1 = \cup_{i=1}^{n} I_{1i}$ and $J_2 = \cup_{i=1}^{n} I_{2j}$ where all I's are disjoint by definition. Then $\mu_{L_1}(J_1 \cup J_2) = \sum_i \mu_{L_0}(I_{1i}) + \sum_j \mu_{L_0}(I_{2j}) = \mu_{L_1}(J_1) + \mu_{L_1}(J_2)$.

Thus extending the μ_{L_0} on $\mathcal{I}$ to finitely additive measure μ_{L_1} on the ring $\mathcal{J}$. □

Our goal now is to extend μ_{L_1} to be a countably additive measure on the σ-algebra.

3 Sigma Algebras and Outer Measures

In the previous lecture, we've shown how to extend the Lebesgue "measure" on the semi-ring of half open, half closed intervals to a finitely additive measure on the on the ring generated by the side of all finite disjoint unions of half open, half closed intervals. In this lecture, we will continue with this reasoning and try to extend our finitely additive Lebesgue measure on our ring to a countable additive measure on a σ algebra which contains our rings, and hence our semi-ring.

3.1 Motivation

Apart from show that it's possible to extend our notion of a "measure" to larger families of sets, you might also wonder what is the purpose of doing so? The reason is this – countably additive measures on σ-algebras allow us to take limits!

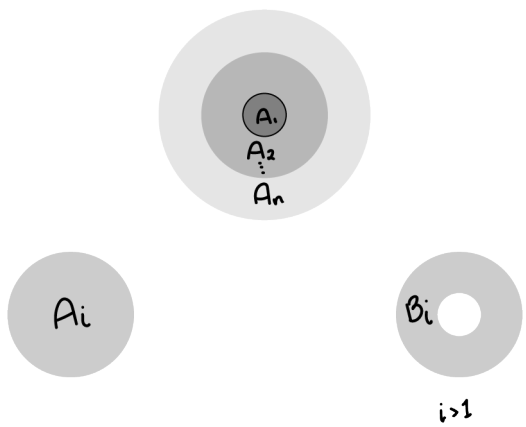
Proposition 3. *Countable additive measures are "continuous" on σ-algebras*

1. $A_i \uparrow A$ $(A_1 \subseteq A_2 \subseteq ...$ *and* $\cup A_i = A) \Rightarrow \mu(A_i) \uparrow \mu(A)$
2. $A_i \downarrow A$ *and* $\mu(A_1) < \infty$ $(A_1 \supseteq A_2 \supseteq ...$ *and* $\cap A_i = A) \Rightarrow \mu(A_i) \downarrow \mu(A)$

Let's look at the proof for (1).

Proof. Define $B_1 = A_1$ and $B_i = A_i \backslash A_{i-1}$ for $i > 1$ as depicted by the figure below.

Figure 3.1: Proposition 3 Part 1



Given how B_i is defined, we know they are disjoint and we know that $A_n = \cup_{i=1}^{n} B_i$, so $A = \cup_i B_i$. By monotonicity, we also know that $\mu(A_{n+1}) \leq \mu(A_n)$.

With what we know, we can say that $\mu(A) = \mu(\cup_i B_i)$ and by countable additivity we can say $\mu(\cup_i B_i) = \sum_i \mu(B_i) = \lim_n \sum_{i=1}^{n} \mu(B_i)$ and now by finite additivity we have $\lim_n \mu(A_n)$. □

3.2 Generating larger families of sets

In the previous lecture, we constructed a ring from a semi-ring. Now, we want to discuss generating *any* larger family of sets which contains some initial family $\mathcal{A} \subseteq 2^{\Omega}$.

For example, we say that ring$(\mathcal{A}) \triangleq$smallest ring that contains $\mathcal{A}$. Similarly for σ-ring($\mathcal{A}$), alg($\mathcal{A}$), or σ-alg($\mathcal{A}$) (which we will just denote by $\sigma(\mathcal{A})$). Slightly more formally, we say that the $*(\mathcal{A}) = \cap\mathcal{B}$ for all sets $\mathcal{B}$ such $\mathcal{A} \subseteq \mathcal{B}$ and $\mathcal{B}$ has the desired properties of $\mathcal{A}$. Given this definition, let's go back to our previous example looking at the set of all half open, half closed intervals on the real line.

Proposition 4. *ring($\mathcal{I}$) $= \mathcal{J} = \{$all finite disjoint union of elements in $\mathcal{I}\}$.*

Proof. Any ring $\mathcal{B}$ that contains $\mathcal{I}$ must also contain all finite disjoint unions. Hence, we know that $\mathcal{J} \subseteq \mathcal{B}$ for all such $\mathcal{B}$. Since we've showed that $\mathcal{J}$ is a ring, this confirms that $\mathcal{J}$ must also be the smallest ring. □

Since our goal for the lecture is to construct a measure on a σ-algebra, naturally we ask what is $\sigma(\mathcal{I})$? Consider the following definition.

Definition 12. Let $\mathcal{X}$ be a topological space (ex: $\mathbb{R}$). The Borel σ-algebra on $\mathcal{X}$ denoted by $\mathcal{B}(\mathcal{X})$ is the smallest σ-algebra that contains all open sets.

Proposition 5. *Let $\mathcal{G} = \{$all open sets in $\mathbb{R}\}$. Then $\sigma(\mathcal{I}) = \sigma(\mathcal{G}) = \mathcal{B}(\mathbb{R})$, i.e., both $\mathcal{I}$ and $\mathcal{G}$ generate $\mathcal{B}(\mathbb{R})$.*

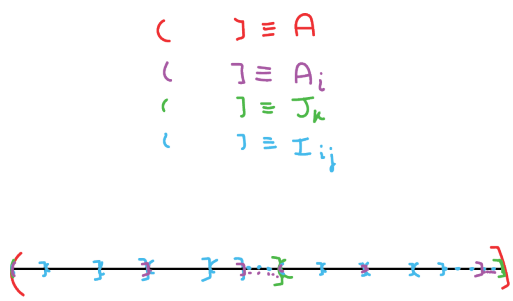
Proof. By definition, $\sigma(\mathcal{G}) = \mathcal{B}(\mathbb{R})$. Now we need to show that $\forall I \in \mathcal{I}$, $I \in \sigma(\mathcal{G})$ and $\forall G \in \mathcal{G}, G \in \sigma(\mathcal{I})$. Since $\mathcal{I}$ is the set of all half-open, half-closed intervals,$I \in \mathcal{I}$ such that $I = (a, b]$, we can say that $(a, b] = \cap_{i=1}^{\infty}\left(a, b + \frac{1}{i}\right) \in \sigma(\mathcal{G})$ since σ algebras are closed under countable intersections. Thus, we know that $\sigma(\mathcal{I}) \subseteq \sigma(\mathcal{G})$. To show that $\sigma(\mathcal{G}) \subseteq \sigma(\mathcal{I})$, we need to use that fact that every nonempty open set in $\mathbb{R}$ is the disjoint union of a countable number of open intervals. Consider $G \in \sigma(\mathcal{G})$ such that $G = (a, b)$. Using that fact, we can say that $(a, b) = \cup_i\left(a, b - \frac{1}{i}\right] \in \sigma(\mathcal{I})$. Thus, we know that $\sigma(\mathcal{G}) \subseteq \sigma(\mathcal{I})$ and can conclude that $\mathcal{B}(\mathbb{R}) = \sigma(\mathcal{G}) = \sigma(\mathcal{I})$. □

Returning to our discussion of the Lebesgue measure, we've already shown that it is finitely additive on the ring($\mathcal{I}$). Now we will show that it is also countably additive on the ring($\mathcal{I}$).

Theorem 4. *μ_{L_1} is countably additive on the ring($\mathcal{I}$)*

Proof. Let $\{A_i\} \subseteq ring(\mathcal{I})$ be countable, and pairwise disjoint and let $A = \cup A_i \in ring(\mathcal{I})$. We want to show that $\mu_{L_1}(A) = \sum_i \mu_{L_1}(A_i)$. The proof for this can be split into steps. The first step is to show that $\mu_{L_1}(A) \geq \sum_i \mu_{L_1}(A_i)$. Then we will show that $\mu_{L_1}(A) \leq \sum_i \mu_{L_1}(A_i)$ which is more involved. For the first step, we know that $\cup_{i=1}^{n} A_i \subseteq A$, so by monotonicity we can say that $\mu_{L_1}(A) \geq \mu_{L_1}(\cup_{i=1}^{n}) = \sum_{i=1}^{n} \mu_{L_1}(A_i)$ by finite additivity on the ring($\mathcal{I}$). Now just let $n \uparrow \infty$ and we're done. For step 2, let's consider the following: let $A = \cup_{k=1}^{K} J_k$ be finite disjoint unions of $J_k \in \mathcal{I}$ and let $\cup_i A_i = \cup_i \cup_{j=1}^{n_i} I_{ij}$ be countable and disjoint since all A_i are disjoint and $I_{ij} \in \mathcal{I}$.

Figure 3.2: Theorem 4 Step 2



By finite additivity, we know that $\mu_{L_1}(A) = \sum_{k=1}^{K} \mu_{L_1}(J_k)$. And again by finite additivity we can say that $\sum_i \mu_{L_1}(A_i) = \sum_i \sum_{j=1}^{n_i} \mu_{L_1}(I_{ij})$. Since our measures are ≥ 0, we can say $\sum_i \sum_{j=1}^{n_i} \mu_{L_1}(I_{ij}) = \sum_{k=1}^{K} \sum_{i,j} \mu_L(I_{ij})$ such that $I_{ij} \subseteq J_k$. Recall that our goal is to show that $\mu_{L_1}(A) \leq \sum_i \mu_{L_1}(A_i) \Rightarrow \sum_{k=1}^{K} \mu_{L_1}(J_k) \leq \sum_{k=1}^{K} \sum_{i,j} \mu_{L_1}(I_{ij})$. If we can show that for some arbitrary J_k that $\mu_{L_1}(J_k) \leq \sum_{i,j} \mu_{L_1}(I_{ij})$ then we will have completed our proof. Let's choose some interval $J = (a, b]$, to restate our goal we want that $\mu_{L_1}((a, b]) \leq \sum_l \mu_{L_1}((a_l, b_l])$ where the intervals $\{(a_l, b_l]\}$ represent a countable, disjoint cover for $(a, b]$. Our goal is to extract a finite cover for the set $(a, b]$. We can say the following,

$$(a + \epsilon, b] \subseteq [a + \epsilon, b] \subseteq (a, b] = \cup_l (a_l, b_l]$$
$$(a_l, b_l] \subseteq \left(a_l, b_l + \frac{\epsilon}{2^l}\right) \subseteq \left(a_l, b_l + \frac{\epsilon}{2^l}\right]$$

which implies that $[a+\epsilon, b] \subseteq \cup_l \left(a_l + b_l + \frac{\epsilon}{2^l}\right)$ and by compactness we know that there must exist a finite cover such that $[a+\epsilon, b] \subseteq \cup_{l\in L} \left(a_l, b_l + \frac{\epsilon}{2^l}\right)$ where L is a finite index set. By the steps above, we now have

$$\begin{aligned}
(a+\epsilon, b) &\subseteq \cup_{l\in L} \left(a_l, b_l + \frac{\epsilon}{2^l}\right) \Rightarrow \\
b-(a+\epsilon) &\leq \sum_{l\in L} \left(b_l + \frac{\epsilon}{2^l} - a_l\right) \text{(monotonicity and finite subadd.)} \\
b-a &\leq \sum_{l\in L} (b_l - a_l) + \sum_{l\in L} \frac{\epsilon}{2^l} + \epsilon \\
b-a &\leq \sum_{l} (b_l - a_l) + 2\epsilon \text{(take } l \text{ to be countable)} \\
\mu_{L_1}((a,b]) &\leq \sum_{l} \mu_{L_1}((a_l, b_l]) \text{ (take } \epsilon \downarrow 0)
\end{aligned}$$

Repeating this process for each J_k, we will have that $\mu_L(A) \leq \sum \mu_l(A_i)$ which implies μ_L is countably additive on ring(I) □

Our new goal, is to extend our notion of μ_{L_1} to the $\sigma(I)$. To do so, we need to develop the idea of an **outer-measure**.

Definition 13. Let $\mu^* : 2^\Omega \to R^+ \cup \{+\infty\}$ be an outer-measure if

1. $\mu^*(\emptyset) = 0$
2. $A \subseteq B \Rightarrow \mu^*(A) \leq \mu^*(B)$
3. If $A \subseteq \cup_i A_i$ is countably and not necessarily disjoint, then $\mu^*(A) \leq \sum \mu^*(A_i)$, i.e. is countable sub-additive

Theorem. *Carathéodory Extension Theorem. Given a countably additive measure, μ, on the ring $\mathcal{A}$. Let*

$$\mathcal{C} = \{A \subseteq \Omega \text{ s.t } \mu^*(E) = \mu^*(E \cap A) + \mu^*(E \cap A^c), \forall E \subseteq \Omega\}$$

call element $A \in \mathcal{C}$ the measurable set.

1. *$\mathcal{A} \subseteq C$ ($\Rightarrow \sigma(\mathcal{A}) \subseteq \mathcal{C}$)*
2. *$\mathcal{C}$ is a σ-algebra*
3. *μ^* is a countably additive measure on $\mathcal{C}$ (and hence on $\sigma(\mathcal{A})$)*
4. *μ^* respects μ on $\mathcal{A}$ ($A \in \mathcal{A} \Rightarrow \mu^*(A) = \mu(A)$).*

Now, let's discuss how to construct the outer-measure. Given a countably additive measure μ on ring $\mathcal{A}$, then $\forall E \subseteq \Omega$ let

$$\mu^*(E) = \inf\left\{\sum_{i=1}^{\infty} \mu(A_i) \text{ s.t. } A_i \text{ are countable cover of } E, A_i \in \mathcal{A}, \cup A_i \supseteq E\right\}$$

If no such cover exists, then we $\mu^*(E) = \infty$

4 Carathéodory's Extension Theorem

In this lecture, we continue our discussion of outer-measures and Carathéodory's extension theorem. Carathéodory's extension theorem is so powerful because it tells us that if we have a measure on a ring then we know there exists a unique extension of that measure to a sigma algebra, so long as certain conditions are met.

4.1 Construction of an outer-measure

Let's recall how we can construct an outer-measure given a countably additive measure μ on a ring $\mathcal{A}$. For all $E \subseteq \Omega$, we have that

$$\mu^*(E) = \inf\left\{\sum_i \mu(A_i) \text{ where } A_i \in \mathcal{A} \text{ countable }, E \subseteq \cup A_i \text{ cover }\right\}$$

and if no such cover exits then $\mu^*(E) = \infty$

Theorem 5. *The construction defined above is, in fact, an outer-measure.*

1. $\mu^*(\emptyset) = 0$
2. $E_1 \subseteq E_2 \subseteq \Omega \Rightarrow \mu^*(E_1) \leq \mu^*(E_2)$
3. $E_{i,}E \subseteq \Omega$ *s.t.* $E \subseteq \cup E_i$ *(countable)* $\Rightarrow \mu^*(E) \leq \sum \mu^*(E_i)$
4. *If* $A \in \mathcal{A}$ *then* $\mu^*(A) = \mu(A)$*, i.e.,* μ^* *respects* μ *on* $\mathcal{A}$*.*

Proof. Let's prove each statements individually.

1. Let each $A_i = \emptyset$, then $\{A_i\}$ is a cover of $\emptyset$ and $\sum_i \mu(\emptyset) = 0$.
2. Any cover of E_2 must also cover E_1.
3. If $\sum \mu^*(E_i) = \infty$ then there is nothing to prove. If $\sum \mu^*(E_i) \leq \infty \Rightarrow \mu^*(E_i) < \infty, \forall i$, however, then we require more. Fix $\epsilon > 0$, choose $A_{ij} \in \mathcal{A}$ s.t. $E_i \subseteq \bigcup_j A_{ij}$, then

$$\mu^*(E_i) \leq \sum_j \mu(A_{ij}) < \mu^*(E_i) + \frac{\epsilon}{2^i}$$

which follows by the definition of the infimum. Since $E \subseteq \bigcup_i E_i \subseteq \bigcup_i \bigcup_j A_{ij}$, then we know that

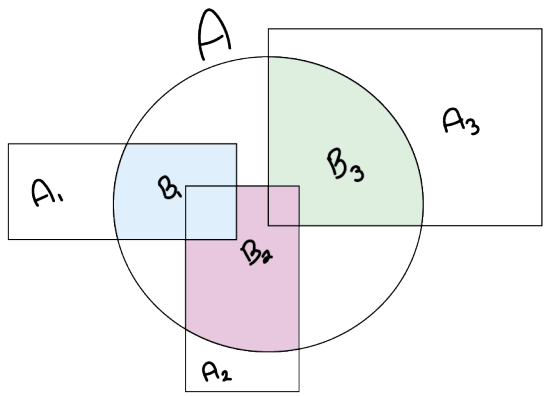
$$\mu^*(E) \leq \sum_{ij} \mu(A_{ij}) \leq \sum_i \left(\mu^*(E_i) + \frac{\epsilon}{2^i}\right) = \sum_i \mu^*(E_i) + \epsilon$$

now let $\epsilon \downarrow 0$, and we have

$$\mu^*(E) \leq \sum_i \mu^*(E_i)$$

4. Let $A \in \mathcal{A}$ show $\mu^*(A) = \mu(A)$. First let's show that $\mu^*(A) \leq \mu(A)$. Let $A_1 = A$, and $A_i = \emptyset$ for $i = 2, 3, ...$ so $\{A_i\}$ is a countable cover of A. This implies that $\mu^*(A) \leq \sum_i \mu(A_i) = \mu(A_1) = \mu(A) \Rightarrow \mu^*(A) \leq \mu(A)$. For the opposite direction, $\mu^*(A) \geq \mu(A)$, we want to show that all covers are $\geq \mu(A)$. Let $\{A_i\} \subseteq \mathcal{A}$ be a countable cover of $\mathcal{A}$, i.e., $A \subseteq \bigcup_i A_i$. Now let $B_i = A \bigcap \left(A_i \backslash \bigcup_{j<i} A_j\right)$ and consider the following diagram below,

Figure 4.1: A and B_i Illustration



In the figure, we did not complete the cover $\{A_i\}$ of A, but we get a general sense for how the B_i are supposed to be interpreted. Once we include the entire cover of A however, notice that we can say that B_i are disjoint, $B_i \in \mathcal{A}$, $B_i \subseteq A_i$ and $A = \bigcup B_i$. This implies that $\mu(A) = \sum \mu(B_i)$ by μ countable additivity on $\mathcal{A}$ and since $A \subseteq \bigcup A_i$, we know by monotonicity that $\mu(A) = \sum \mu(B_i) \leq \sum_i \mu(A_i)$ which generalizes to be true for all covers $\{A_i\}$ thus accomplishing our goal in showing that $\mu(A) \leq \mu^*(A)$ and concluding our proof that $\mu(A) = \mu^*(A)$. $\square$

Our goal for this section is to find a σ-algebra, $\mathcal{F} \subseteq 2^\Omega$ s.t. $\mathcal{A} \subseteq \mathcal{F}$ and μ^*, the outer measure defined in 2^Ω, is a countably additive measure on $\mathcal{F}$.

4.2 Carathéodory's Extension Theorem

Given a countably additive measure μ defined on the ring $\mathcal{A}$. Let μ^* be the outer measure that respects μ on $\mathcal{A}$. Now let

$$\mathcal{C} = \{A \subseteq \Omega \text{ s.t } \mu^*(E) = \mu^*(E \cap A) + \mu^*(E \cap A^c), \forall E \subseteq \Omega\}$$

and call sets in $\mathcal{C}$ the **measurable sets.** Then,

1. $\mathcal{A} \subseteq \mathcal{C}$
2. $\mathcal{C}$ is a σ-alg
3. μ^* is a countably additive measure on $\mathcal{C}$ and hence on $\mathcal{A}$

Let's make a few observations regarding the statement. The first is that (1) and (2) imply that $\mathcal{A} \subseteq \sigma(\mathcal{C}) = \mathcal{C} \Rightarrow \sigma(\mathcal{A}) \subseteq \mathcal{C}$. This means that $\mathcal{C}$ is a σ-algebra that contains $\sigma(\mathcal{A})$, but it's not necessarily the smallest. We also know that by countable sub-additivity of μ^*, we must always have that when $E \subseteq \cup E_i \Rightarrow \mu^*(E) \leq \sum \mu^*(E_i)$. In this example, we can just let $E_1 = E \cap A$ and $E_2 = E \cap A^c$ and we have that $\mu^*(E) \leq \mu^*(E \cap A) + \mu^*(E \cap A^c)$. Therefore, the difficulty arrises in relationship to the Carathéodory Extension Theorem when showing equality.

Let's consider the following example. Let $A \bigcap B = \emptyset$ and let $A \in \mathcal{C}$, so we know it's a measurable set. Then we can say the following

$$\begin{aligned} \mu^*(A \cup B) &= \mu^*((A \cup B) \cap A) + \mu^*((A \cup B) \cap A^c) \\ &\mu^*(A) + \mu^*(B) \end{aligned}$$

i.e., they are equal in subadditivity.

Now, we will attempt to cover the intuition for splitting. Recall for the Riemann integral, the goal was to take the limit such that we the upper bound equal to the lower bound. Similarly, we consider the outer

measure, $\mu^*(A)$ and the "inner" measure which we'll define by

$$\mu_*(A) = \mu^*(\Omega) - \mu^*(A^c)$$

If these two objects are equal, then we'll have $\mu^*(\Omega) = \mu^*(A) + \mu^*(A^c)$ as desired. Since, there's nothing inerintly special about Ω, we can split it up with respect to any $E \subseteq \Omega$ such that

$$\mu^*(E) = \mu^*(E \cap A) + \mu^*(E \cap A^c)$$

Recall that by the very definition of the outer-measure, we are given subadditivity for free ($\leq$), but now we need to prove ($\geq$). Let assume that $\mu^*(E) < \infty$ or else the proof is trivial.

Proof. Proof of Carthéodory's Extension Theorem. We first need to show that $\mathcal{A} \subseteq \mathcal{C}$, i.e., if $A \in \mathcal{A}$ then $A \in \mathcal{C}$.

$$\mu^*(E) = \mu^*(E \cap A) + \mu^*(E \cap A^c), \forall E \subseteq \Omega$$

We already know that ($\leq$) is given by subadditivity so we need to go about showing ($\geq$). Let's fix $E \subseteq \Omega$ such that $\mu^*(E) < \infty$ and fix $\varepsilon > 0$. Take $\{A_i\} \subseteq \mathcal{A}$ be a ocuntable cover of E such that (1) $E \subseteq \cup_i A_i$ and (2)$\mu^*(E) \leq \sum \mu(A_i) < \mu^*(E) + \varepsilon$. We just need to show that the cover can be used to cover $E \cap A$ and $E \cap A^c$. We know that $\{A_i\}$ is a countable cover for E, therefore we know that $E \cap A \subseteq \cup_i (A_i \cap A)$ and $E \cap A^c \subseteq \cup_i (A_i \cap A^c)$. We can also say that $A_i \cap A$ and $A_i \cap A^c$ belong to the ring $\mathcal{A}$ where recall $\mu(A)$ is a defined as a countably additive measure. Now we can take all of our pieces and say that

$$\begin{aligned}
\mu^*(E \cap A) + \mu^*(E \cap A^c) &\leq \sum_i [\mu(A_i \cap A) + \mu(A_i \cap A^c)] \\
&= \sum_i \mu(A_i) \\
&\leq \mu^*(E) + \varepsilon \text{ (by our definition)}
\end{aligned}$$

Now just let $\varepsilon \downarrow 0$ and we finally have that $\mu^*(E \cap A) + \mu^*(E \cap A^c) \leq \mu^*(E)$ thus completing our proof for $\mu^*(E) = \mu^*(E \cap A) + \mu^*(E \cap A^c)$ and show that $\forall A \in \mathcal{A} \Rightarrow A \in \mathcal{C}$. Now we must show that $\mathcal{C}$ is a σ-algebra. We will now check that each conidtion is satisfied. (1) $\emptyset \in \mathcal{C}$. Let $A = \emptyset$, then $\mu^*(E) = \mu^*(\emptyset) + \mu^*(E \cap \Omega) = \mu^*(E)$. (2) If $A \in \mathcal{C}$ then $A^c \in \mathcal{C}$. This is true by the symmetric definition of the measurable sets. If $A, B \in \mathcal{C}$ then $A \cap B \in \mathcal{C}$. We know that $A \in \mathcal{C}$, so we have that

$$\begin{aligned}
\mu^*(E) &= \mu^*(E \cap A) + \mu^*(E \cap A^c) \\
&= \mu^*((E \cap A) \cap B) + \mu^*((E \cap A) \cap B^c) + \mu^*(E \cap A^c) \text{ (since we know } B \in \mathcal{C}) \\
&= \mu^*(E \cap (A \cap B)) + \mu^*(E \cap (A \cap B)^c \cap A) + \mu^*(E \cap (A \cap B)^c \cap A^c) \\
&= \mu^*(E \cap (A \cap B)) + \mu^*(E \cap (A \cap B)^c)
\end{aligned}$$

thus showing that $A \cap B \in \mathcal{C}$. And since $A \cap B = (A^c \cap B^c)^c$, then we know that $\mathcal{C}$ is closed under complements so it must be an algebra. To show that it's a σ-algebra, we need to verify that $\mathcal{C}$ is also closed under countable unions. Let's consider the set $\{A_i\} \subseteq \mathcal{C}$ and let $B = \cup A_i$ such that $B \in \mathcal{C}$. Define $B_n \triangleq \cup_{i=1}^n A_i \in \mathcal{C}$ by finite $\cup$. WLOG, assume that A_i are disjoint, thus B_n is a disjoint union of $A_1, ..., A_n$. We now have that

$$\begin{aligned}
\mu^*(E) &= \mu^*(E \cap B_n^c) + \mu^*(E \cap B_n) \\
&= \mu^*(E \cap B_n^c) + \mu^*(E \cap B_n \cap A_n) + \mu^*(E \cap B_n + A_n^c) \\
&= \mu^*(E \cap B_n^c) + \mu^*(E \cap A_n) + \mu^*(E \cap B_{n-1})
\end{aligned}$$

by recursion, this impliwa that

$$\mu^*(E \cap B_n) = \mu^*(E \cap A_n) + \mu^*(E \cap B_{n-1})$$

$$
\begin{aligned}
&= \mu^*\left(E \cap B_n^c\right) + \sum_{i=1}^{n} \mu^*\left(E \cap A_i\right) \\
&\geq \mu^*\left(E \cap B^c\right) + \sum_{i=1}^{n} \mu^*\left(E \cap A_n\right) \\
\mu^*\left(E\right) &\geq \mu^*\left(E \cap B^c\right) + \sum_{i} \mu^*\left(E \cap A_n\right) \\
&\geq \mu^*\left(E \cap B^c\right) + \mu^*\left(E \cap B\right)
\end{aligned}
$$

and we have $\leq$ by subadditivity so we know that countable unions are in $\mathcal{C}$.

Now, we want to show that μ^* is countably additive on the σ-algebra $\mathcal{C}$. Let $\{A_i\} \subseteq \mathcal{C}$ be disjoint, countable. Let $B = \cup A_i \in \mathcal{C}$. We have that $\mu^*\left(B\right) \leq \sum \mu^*\left(A\right)$ by countable subadditivity but now we want that $\mu^*\left(B\right) \geq \sum \mu^*\left(A\right)$. We know from the previous proof that $\mu^*\left(E\right) \geq \mu^*\left(E \cap B^c\right) + \sum_i \mu^*\left(E \cap A_i\right)$. Now let $E = B$ and we have

$$
\begin{aligned}
\mu^*\left(B\right) &\geq 0 + \sum_{i} \mu^*\left(B \cap A_i\right) \\
&= \sum_{i} \mu^*\left(A_i\right)
\end{aligned}
$$

thus completing our proof. □

We have finally constructed a countably additive measure on a σ-algebra based on a countably additive measure on a ring.

4.3 Existence and Uniqueness

Now, we want to ask under what conditions the measures we've constructed on our σ-algebra exist and unique. Let μ be a countably additive measure on the algebra $\mathcal{A}$. Let λ be a countably additive measure on the $\sigma\left(\mathcal{A}\right)$ that respects μ on $\mathcal{A}$. We actually already know that λ exists by the Caratheadoy Theorem, but is it unique? The answe is no!

Consider the following counter example. Let $\mathcal{A} = \operatorname{alg}\left(\mathcal{I}\right)$ and let $\mu\left(A\right) = +\infty$ whenever $A \neq \emptyset$. Now, let's define $\mathcal{B} = \sigma\left(\mathcal{A}\right)$. For $B \in \mathcal{B}$ let $\lambda_1\left(B\right) = +\infty$ whenever $B \neq \emptyset$. Note that λ_1 respects μ on $\mathcal{A}$. But now also let $\lambda_2\left(B\right) =$ counting measure. Note, $\lambda_2\left(B\right)$ also respects μ on $\mathcal{A}$ because counting the number of elements in an interval is uncountably infinite. However, notice the following difference when considering singletons

$$
\begin{aligned}
\lambda_1\left(\{x\}\right) &= +\infty \\
\lambda_2\left(\{x\}\right) &= 1
\end{aligned}
$$

Why does this happen? It's because the algebra of intervals actually does not contain singletons. In the section, we will address how to fix this problem.